# Run report: ctrlK-q3-nonsplit-run-20261024T2153Z

Descriptive record of one recorded run of examples/genus4_n4_census.rs, part CENSUS (exhaustive, q = 3) under `research/genus4_n4_census_20261024/PROTOCOL.md` (with its dated amendments). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations ($F_p$ multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, or a RunPod CPU pod whose lscpu is in host.txt); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

## env.txt

```
run: ctrlK-q3-nonsplit 20261024T2153Z
git HEAD: 0fb870d0ee76ba6a1b059e9e34c869518fb412b0
tool: examples/genus4_n4_census.rs sha256 f4c7f7355c838b6f112e09cb412fd3353d087309962c64cd39ad49d79f4c9dad
protocol: research/genus4_n4_census_20261024/PROTOCOL.md sha256 a95b1e35748cefe23a044d8cfd9e059b723cf71fdbc4af56db71257da42c8b33
command: /tmp/claude-0/g4/g4 research/genus4_n4_census_20261024/results/ctrlK-q3-nonsplit-run-20261024T2153Z CENSUS 3 nonsplit exh 1 0 2000
rustc: rustc 1.97.0 (2d8144b78 2026-07-07)
host: Linux 6.18.44-fc-v153 x86_64, 4 cores, Intel(R) Xeon(R) Processor @ 2.10GHz
threads: 1
wall seconds: 196
```

## tool_log.txt

```
nonsplit 3: 26891264/43046721 samples, res-shaped 170092, allowed-trace cand 19023, smooth hits 1660, 120s
nonsplit q=3: samples=43046721 (exhaustive) res_shaped=274380 allowed_trace_candidates=31680 smooth_prime_order_hits=2880 smooth_fraction_among_prime_cands=0.0909
smooth_fraction_other_res_shaped(sub-sample 2000)=196/2000 secs=195.7
nonsplit q=3: total prime-order mass estimate 1.000 +- 0.019 (hit rate per sampled form 6.690e-5)
```

## summary.txt

```
nonsplit q=3: samples=43046721 (exhaustive) res_shaped=274380 allowed_trace_candidates=31680 smooth_prime_order_hits=2880 smooth_fraction_among_prime_cands=0.0909
smooth_fraction_other_res_shaped(sub-sample 2000)=196/2000 secs=195.7
nonsplit q=3: total prime-order mass estimate 1.000 +- 0.019 (hit rate per sampled form 6.690e-5)
```

## cand_t_hist.csv (first 12 of 18 rows)

| type | t | candidates |
|---|---|---|
| nonsplit | −80 | 2160 |
| nonsplit | −17 | 1440 |
| nonsplit | −16 | 5760 |
| nonsplit | −14 | 5760 |
| nonsplit | −13 | 11520 |
| nonsplit | −12 | 2880 |
| nonsplit | −11 | 8640 |
| nonsplit | −10 | 5760 |
| nonsplit | −9 | 14400 |
| nonsplit | −8 | 50400 |
| nonsplit | −6 | 14400 |
| nonsplit | −5 | 8640 |

## hits.csv (first 12 of 31680 rows)

| type | code | t | nE | smooth |
|---|---|---|---|---|
| nonsplit | 10196 | −1 | 83 | false |
| nonsplit | 10660 | −1 | 83 | false |
| nonsplit | 12140 | −1 | 83 | false |
| nonsplit | 13090 | −1 | 83 | false |

| | | | | |
|---|---|---|---|---|
| nonsplit | 16433 | −1 | 83 | false |
| nonsplit | 17383 | −1 | 83 | false |
| nonsplit | 18863 | −1 | 83 | false |
| nonsplit | 19327 | −1 | 83 | false |
| nonsplit | 23053 | −1 | 83 | false |
| nonsplit | 23723 | −1 | 83 | false |
| nonsplit | 25159 | −1 | 83 | false |
| nonsplit | 25991 | −1 | 83 | false |

### mass_by_t.csv (first 1 of 1 rows)

| type | t | nE | smooth_hits | mass_estimate | mass_stderr |
|---|---|---|---|---|---|
| nonsplit | −1 | 83 | 2880 | 1.0000 | 0.0186 |

### Graph

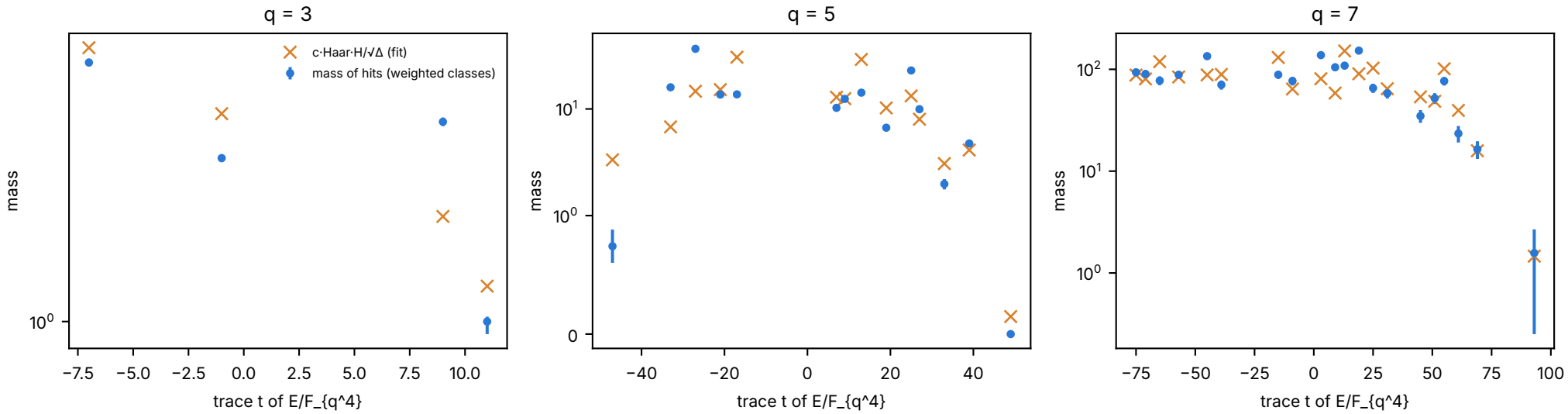



Figure 1: Mass of hits per prime-order trace with the Haar-density fit.